

Technical Report of CoachAI Challenge 2023: Team NTNUEE AIoTLab

Po-Yung Chou , Yu-Chun Lo , Bo-Zheng Xie , Cheng-Hung Lin* and Yu-Yung Kao

National Taiwan Normal University
corresponding author: *brucelin@ntnu.edu.tw

Abstract

The CoachAI Badminton 2023 Track1 initiative aim to automatically detect events within badminton match videos. Detecting small objects, especially the shuttlecock, is of quite importance and demands high precision within the challenge. Such detection is crucial for tasks like hit count, hitting time, and hitting location. However, even after revising the well-regarded shuttlecock detecting model, TrackNet, our object detection models still fall short of the desired accuracy. To address this issue, we've implemented various deep learning methods to tackle the problems arising from noisy detected data, leveraging diverse data types to improve precision. In this report, we detail the detection model modifications we've made and our approach to the 11 tasks. Notably, our system garnered a score of 0.78 out of 1.0 in the challenge.

1 Introduction

The objectives of CoachAI Badminton 2023 Track1 [Ik *et al.*, 2023] are described as follows. Participants are required to construct a computer vision system that generates 11 targets from an input video. For clarity, the 11 tasks are listed below:

Number of shots The total number of shots within a match, which is equivalent to the number of hits. It's crucial to note that if the predicted number of shots deviates from the ground truth, the task will be scored as 0 points.

Hit frame This denotes the precise moment of hitting in the video. Any prediction error exceeding 2 frames will be deemed incorrect.

Hitte The task is to identify which player strikes the shuttlecock.

BallHeight This refers to the altitude of the hitting point.

RoundHead Classify whether the player's action was a round head shot or not during the hit.

Backhand Determine if the player used a backhand action during the hit.

LandingX/LandingY This represents the shuttlecock's projected position on the court at the end of its trajectory,

whether struck by the opponent or when it touches the ground, measured in pixels.

HitteLocationX/HitteLocationY This signifies the position of the player executing the strike, using the toe of their lead foot as the coordinate reference in the image, in pixels. If the player jumps, the projected coordinates onto the court are considered.

DefenderLocationX/DefenderLocationY This denotes the position of the player on the defensive, and is determined in a manner similar to the hitting player's location.

BallType There are nine types of balls, labeled from 1 to 9. For additional details, refer to: <https://sites.google.com/view/coachai-challenge-2023/tasks/track1>

Winner This task involves identifying the winning player.

To construct a system capable of addressing all these tasks, we've adopted a two-step process. The first and foundational function for these tasks is the object detection model, which is associated with aspects like location projection, the trajectory of the object, and the object's precise location, among others. After determining the object's location, a post-processing model is employed to produce the predictions for different task. In the subsequent two sections, we will delve into the object detection techniques in Section 2 and describe the methods we've implemented for each task in Section 3.

2 Object Detection

The object detection model employed in this report primarily provides information regarding the sequence of the shuttlecock and the location of the players. Given that location in a badminton game is relative to the court and the net, the model also needs to detect these two elements. Additionally, the racket is detected to ensure the measurement of the habit foot. The five types of objects we aim to detect using the model are: **shuttlecock**, **player**, **court**, **net**, and **racket**.

We have chosen YOLOv7 [Wang *et al.*, 2023] as our object detection model due to its impressive performance on the COCO dataset [Lin *et al.*, 2014]. Based on YOLOv7, the player, court, and net can be predicted with precision. While the racket is challenging to detect, the model still provides sufficient results to allow us to analyze which foot is the habitual one. However, detecting the shuttlecock is problematic

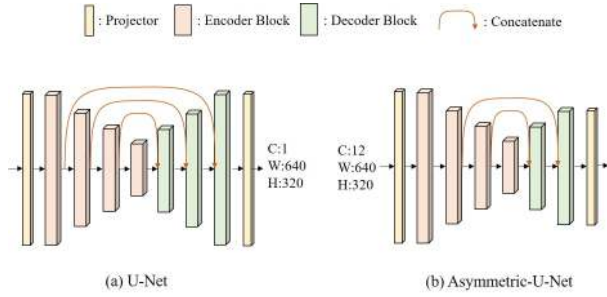


Figure 1: U-Net and Asymmetric U-Net

due to its fast movement, which results in a blur effect, and its relatively small size in the frame.

Fortunately, TrackNet[Huang *et al.*, 2019] was introduced to detect objects such as the shuttlecock and tennis ball, both of which move rapidly and are small on-screen. During the training phase, we discovered that to detect these objects more precisely, a larger input size might be beneficial. Yet, this increase in size subsequently leads to longer training durations and higher computational demands. Most notably, when we consider using the maximum input resolution, equivalent to the original video dimensions of 1280×720 , the model becomes untrainable, even with a batch size set to 1.

To improve the detection results for the shuttlecock, we decided to modify the original U-Net structure used in TrackNet to an Asymmetric U-Net, as illustrated in Fig.1. Furthermore, to allow the model to learn features at different resolutions, we use a ground-truth heatmap to supervise outputs from every decoder block. The F1-score comparison is provided in Tab.1, and the computation cost comparison can be found in Tab.2. Experiments were conducted on a dataset we collected, comprising 128 training videos and 17 testing videos. Each video has frame-by-frame labeling.

Table 1: The Comparison of F1-score on different model with input resolution 640×640

Method	F1-score
U-Net	0.93866
Asym U-Net	0.94255

Table 2: The Comparison of FLOPs on different model with input resolution 640×640

Method	FLOPs(G)
U-Net	255.8
Asym U-Net	188.26

After predicting the shuttlecock’s location throughout the entire video, we implemented a denoising process. This process involves removing “jumping” points using a median filter. Additionally, we filtered out points from the game’s start and end using a straightforward timeline. We can reasonably assume the shuttlecock remains static during these pe-

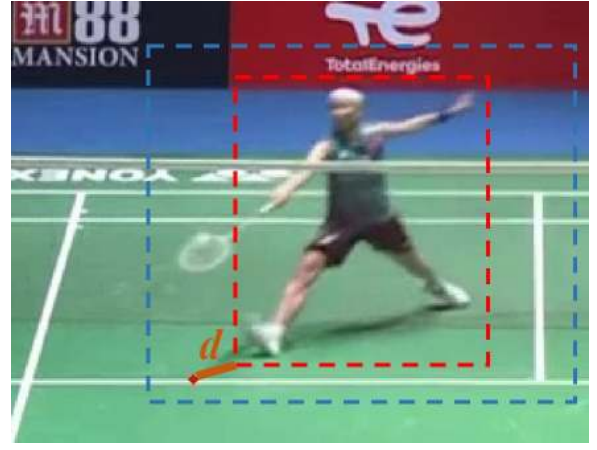


Figure 2: The illustration of input data for the “Location” prediction.

riods, given that each video captures only one game. After the removal process, we applied interpolation to reconstruct the shuttlecock’s trajectory.

Up to this point, we have a set of shuttlecock locations. However, these results are not yet refined enough to directly make accurate predictions regarding ball type, hit number, and so on. Therefore, we need an additional post-processing model.

Let me add a supplementary note on the habit foot: we need to use VitPose[Xu *et al.*, 2022] to predict human pose and combine it with the result of racket detection to determine habit foot.

3 Post-process Methods

3.1 Number of shots, Hit frame and Hitter

The most straightforward method to detect the hitting time involves counting the inverse numbers of the moving direction on the y-axis. However, noise from the shuttlecock detections can easily lead to errors in these numbers. Consequently, we opted for an alternative approach: predicting whether “Player A hits”, “Player B hits”, or from a given clip. This clip is generated using a sliding window of size 4 that moves across the badminton video. For this task, we employed the X3D-M[Felichtenhofer, 2020] model, pretrained on the kinetics-400[Kay *et al.*, 2017] dataset, to encode the spatiotemporal features necessary for these predictions.

The labels for training the video analysis model are derived directly from the ground truth. For instance, if the 5th frame is marked in the ground truth as the frame where Player A hits, then the clips spanning frames $2 \rightarrow 6$, $3 \rightarrow 7$, and $4 \rightarrow 8$ are automatically labeled as “Player A hits”.

After the X3D model produces its results, a sequence of predictions emerges. We then apply a smoothing filter to this sequence and select the most frequent category as the final prediction, yielding a sequence like [None, None, A, A, A, None, None]. In this context, None represents “nothing happens”, A stands for “Player A hits”, and B indicates “Player B hits”. From the example sequence, the instance of “Player A hits” clearly occurs at the third frame.

Using this method, we can get results for the task **Number of shots**, **Hit frame**, and **Hitter**.

3.2 BallHeight, LandingX/LandingY, HitterLocationX/HitterLocationY, DefenderLocationX/DefenderLocationY

To predict ball height, the intuitive approach involves using the shuttlecock location at the hit frame and then mapping this location to the foot’s y-axis. However, this method doesn’t perform well due to noise in both the shuttlecock detection and human pose estimation. To address this, we employ EfficientNet[Tan and Le, 2019] to predict the pixel offsets, denoted as dx and dy . These values, dx and dy , are determined by the “CORNER” and the ground truth. The term “CORNER” refers to the lower corner of the player’s bounding box that is closest to the shuttlecock’s location.

The automatic labeling process is depicted in Fig.2. The red line represents the output from YOLOv7, while the blue line outlines a bounding box with dimensions $W \times 1.8$ and $H \times 1.4$. We feed the region enclosed by the blue box into EfficientNet to predict the offsets dx and dy . We use this blue area because it encompasses both the shuttlecock and the projection point of the ground truth.

The method can be implemented for tasks **BallHeight**, **LandingX/LandingY**, **HitterLocationX/HitterLocationY**, and **DefenderLocationX/DefenderLocationY**.

It’s worth noting that we have tried various methods to determine dx and dy , such as the difference between the ground truth projection point and the shuttlecock, and the difference between the ground truth projection point and the player’s center, etc. However, the most accurate and consistent results were achieved using the method mentioned earlier in this section.

3.3 Backhand

We directly use X3D to predict whether the player’s action is a backhand or not. The training process is similar to that described in Section 3.1. We chose not to rely on human pose estimations for making predictions due to noise issues; VitPose cannot yield accurate results when the player is too blurry in the video.

3.4 RoundHead

The task of identifying a “RoundHead” action is more challenging than “Backhand”. This complexity arises because the action can vary significantly, and its appearance may differ depending on the player’s position relative to the camera. To mitigate this, we integrate sequences of the shuttlecock’s location, the player’s location, the court’s location, and the net’s location. As illustrated in Fig.3, we feed these sequences into a transformer[Vaswani *et al.*, 2017] to extract location features. The insights derived from these location sequences are valuable; for instance, the relative positions of the player and the shuttlecock can influence the player’s hitting pose. Notably, we employ a hint vector: “Player A” corresponds to $[1,0]$ and “Player B” to $[0,1]$. This vector influences the X3D’s output, as highlighted in Fig.3 by the red text. We use a single fully connected layer to project this hint vector to match the feature dimensions.

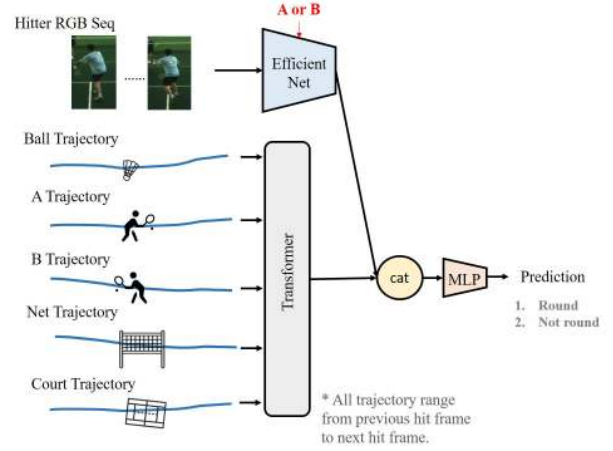


Figure 3: The model of RoundHead task



Figure 4: The RoI for the input of the winner model

3.5 BallType

We integrate sequences of the player’s location, the court’s location, and the net’s location for prediction. We opted not to use the shuttlecock location sequence due to its inherent noise. For this task, the transformer doesn’t use a mask to adapt to different input lengths. Instead, we set absent input tokens to -100 , ensuring that fast shots and long shots have distinct input value ranges (fast shots should be smaller than long shots).

3.6 Winner

Directly using the court boundary and shuttlecock location can yield inaccurate results. We’ve previously discussed the inaccuracy of the shuttlecock’s location. The potential inaccuracies in the court boundaries can be attributed to lighting conditions and camera angles. Therefore, we utilize the sequence from the last several RGB frames to determine the winner. To ensure the model remains focused and doesn’t get sidetracked by extraneous details from the entire frame, we select the net area and the court’s four boundary areas, as depicted in Fig.4, for input. This is analogous to having five cameras monitoring the game.

3.7 Demoenstration Video

We have prepared some demonstration videos, available at the following URL:

<https://www.youtube.com/shorts/BsOyQM44f28>

4 Conclusion

Benefiting from the wealth of annotated data provided by CoachAI Badminton 2023 Track1, we can employ a “deeper” learning strategy to address the inherent challenges. Our experiments demonstrated that, to the human eye, object detection results may seem satisfactory at first glance, but they don’t always achieve the level of accuracy necessary for tasks focused on exact location determination. This highlights the crucial role of advanced deep learning algorithmic techniques in narrowing the disparity between human observational aptitudes and the exacting standards of our analytical tasks. As we look to the future, AI coaching systems hold the promise of numerous groundbreaking applications, poised to make a significant impact on our daily lives.

References

- [Feichtenhofer, 2020] Christoph Feichtenhofer. X3d: Expanding architectures for efficient video recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020.
- [Huang *et al.*, 2019] Yu-Chuan Huang, I-No Liao, Ching-Hsuan Chen, Tsi-Uí Ik, and Wen-Chih Peng. Tracknet: A deep learning network for tracking high-speed and tiny objects in sports applications. In *2019 16th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, pages 1–8, 2019.
- [Ik *et al.*, 2023] Tsi-Ui Ik, Wen-Chih Peng, Jiun-Long Huang, Wei-Yao Wang, Wei-Wei Du, Guan-Yi Jhang, Yung-Chang Huang, and Jiun-Chi Wang. Coachai badminton challenge 2023. <https://sites.google.com/view/coachai-challenge-2023/tasks/track1>, 2023.
- [Kay *et al.*, 2017] Will Kay, João Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, Mustafa Suleyman, and Andrew Zisserman. The kinetics human action video dataset. *CoRR*, abs/1705.06950, 2017.
- [Lin *et al.*, 2014] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft coco: Common objects in context. In David Fleet, Tomas Pajdla, Bernt Schiele, and Tinne Tuytelaars, editors, *Computer Vision – ECCV 2014*, pages 740–755, Cham, 2014. Springer International Publishing.
- [Tan and Le, 2019] Mingxing Tan and Quoc Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 6105–6114. PMLR, 09–15 Jun 2019.
- [Vaswani *et al.*, 2017] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.
- [Wang *et al.*, 2023] Chien-Yao Wang, Alexey Bochkovskiy, and Hong-Yuan Mark Liao. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7464–7475, June 2023.
- [Xu *et al.*, 2022] Yufei Xu, Jing Zhang, Qiming ZHANG, and Dacheng Tao. Advances in neural information processing systems. volume 35, pages 38571–38584. Curran Associates, Inc., 2022.